

RFID Long Range Reader

Extended Range 125 kHz Reader using RDM63000 and Custom
Hand-Wound Coil

Student Project Report — Computer and Network Security

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Abstract

When we first started looking into 125 kHz RFID systems, one thing stood out immediately: modules like the RDM63000 are cheap and easy to find, but their built-in coils barely manage a read range of 3 cm. That kind of distance is fine for tapping a card on a reader right in front of you, but it rules out almost every practical scenario where you actually want contactless identification to feel effortless. This report tells the story of what we did about it.

Our approach was to replace the small stock coil with a custom hand-wound coil of much larger diameter, and to tune the resulting LC circuit so that it resonates at exactly 125 kHz. To get there, we first had to abandon our original plan. We started with the MFRC522, a popular 13.56 MHz module, only to discover that its coil is printed directly onto the PCB and simply cannot be swapped out. Switching to the 125 kHz RDM63000, which is built from the ground up to accept an external coil, opened the door to the experiments that followed.

The honest result of those experiments was a consistent read range of approximately 3 cm across all configurations tested — no better than the stock module. The project's real contribution is the systematic identification of why: the RDM63000's internal driver limits coil output to approximately 1.4 V regardless of external supply voltage, providing insufficient current to power a passive tag beyond 3–4 cm. Multiple attempts to overcome this through resistor voltage dividers, battery bridges, and varied coil wire gauges each failed for specific, calculable reasons documented in this report. The theoretical framework — LC resonance, Q-factor optimisation, Biot–Savart field calculations — was validated throughout; it was the available drive power, not the antenna design, that set the ceiling.

1 Introduction and Motivation

1.1 Problem Statement

When we began this project, our first instinct was to reach for the MFRC522 — the 13.56 MHz RFID module that appears in practically every Arduino tutorial online. It seemed like the obvious starting point. But we hit a wall almost immediately: the MFRC522's antenna coil is not a separate component you can unplug and replace. It is etched directly into the PCB as a printed trace, and the driver circuitry is tuned to that specific geometry. Replacing it would mean cutting into the board itself, which is not a realistic option.

That obstacle forced us to rethink the whole approach. We ended up switching to the **125 kHz RDM63000 module**, which is designed with external coil support in mind. Its coil connection pins are fully accessible, meaning we could disconnect the stock coil entirely and wire up our own. That single change unlocked everything else — the ability to experiment with coil size, turn count, and tuning capacitors in a way that simply was not possible with the MFRC522.

1.2 Project Goals

1. Understand concretely why the MFRC522 was not suitable, and explain what made the switch to a 125 kHz system worthwhile.
2. Build enough of a physics foundation to make informed decisions about coil geometry and resonance tuning.
3. Design and wind a large-diameter loop antenna that resonates at 125 kHz, calculated using the LC resonance formula.
4. Connect the custom coil to the RDM63000 and tune the LC circuit by selecting appropriate capacitor values.
5. Power the whole system from an Arduino Mega at 5 V and achieve reliable tag reads at the greatest distance possible.
6. Document every step — including the things that did not work — clearly enough that someone else could replicate or improve on our results.

2 Theoretical Background

2.1 125 kHz RFID and EM4100/HID Communication

125 kHz RFID has been around for decades, and there is a good reason it has never gone away: it works reliably, requires no battery in the tag, and penetrates most everyday materials without much trouble. The way it works is straightforward. The reader continuously generates an alternating magnetic field at 125 kHz. When a passive tag enters that field, its internal coil picks up enough energy to power the chip, which then responds by slightly modulating the field — either through amplitude-shift keying (ASK) or frequency-shift keying (FSK). The reader detects those tiny variations and decodes the tag’s ID. No contact, no battery, no problem.

What determines how far away a tag can be read is almost entirely the strength of the magnetic field at that distance. Magnetic field strength from a loop antenna falls off roughly with the cube of the distance, which sounds discouraging, but it also means that anything you can do to increase the field strength at the source — bigger coil, better tuning — pays off quickly. Compared to 13.56 MHz systems, the lower frequency of 125 kHz brings better penetration through walls and materials, though it does demand physically larger coils to generate the same level of inductance.

2.2 Resonance and the LC Circuit

The key insight behind this whole project is that a coil alone is not enough. What you really want is an LC resonant circuit — a coil (inductor) paired with a capacitor — because at the resonant frequency the two components cancel each other’s reactance, leaving only the coil’s small DC resistance in the path. At that point, the circulating current in the tank circuit, and therefore the magnetic field it generates, reaches a sharp peak. Everything else being equal, a well-tuned LC circuit will produce dramatically more field than a bare coil driven at the same voltage.

The resonant frequency is:

$$f = \frac{1}{2\pi\sqrt{LC}} \quad (2.1)$$

where L is the coil inductance in henries and C is the capacitance in farads. To hit our target of 125 kHz, we rearrange for the capacitance needed:

$$C = \frac{1}{(2\pi f)^2 L} \quad (2.2)$$

As a concrete example: a 14 cm coil wound to 48 turns can reach an inductance around

1 mH, and plugging that into equation (2.2) gives a required tuning capacitance of roughly 1.6 nF — a perfectly standard value that is easy to source.

2.3 Why 125 kHz Supports External Coils

The practical reason we could even attempt this project comes down to how 125 kHz modules are engineered. Unlike the MFRC522, where the coil is baked into the PCB and the driver is matched to it at the factory, the RDM63000 and similar low-frequency modules expose their coil pins openly. The internal oscillator is built to drive whatever LC tank you connect to it. As long as your coil and capacitor together resonate near 125 kHz, the module will excite the circuit and the field will build up naturally. No additional matching network is required. That accessibility is exactly what made our custom coil experiment feasible.

3 Hardware Design

3.1 Component List

Component	Specification	Qty
RDM63000 module	125 kHz, external coil pins, 5 V supply	1
Arduino Mega	ATmega2560, 5 V logic, multiple UART ports	1
Enameled copper wire	Magnet wire, insulating varnish coating	1
Capacitors	Various values for LC tuning at 125 kHz	5 ea.
Breadboard	830-tie full size	1
Ammeter	For coil continuity verification	1
Multimeter	For voltage and resistance checks	1
LED (green/red)	5 mm through-hole	2
Resistors	220 Ω for LED current limiting	2
125 kHz RFID tags	EM4100-compatible passive tags	3

Table 3.1. Full bill of materials for the 125 kHz RFID extended-range system.

3.2 Coil Design and Construction

We built and tested four different coil configurations using enameled magnet wire. Two of the coils were circular, while the other two were rectangular. These different geometries were investigated to study how coil shape, dimensions, and number of turns influence the inductance and resonance behavior at 125 kHz.

Coil A was wound as a circular loop with a diameter of 14 cm and only 3 turns. This produces a very low inductance, meaning that the capacitance required to achieve resonance at 125 kHz would be extremely large and outside typical standard capacitor values. It was nevertheless constructed as a baseline reference for comparison.

Coil B was also circular with the same 14 cm diameter, but with 48 turns. The higher number of turns increases the inductance significantly, bringing the required tuning capacitance into a practical range for resonance at 125 kHz.

Coil C was constructed as a rectangular loop with dimensions of 11.5 cm $\times$ 7.5 cm and 50 turns. The rectangular geometry was tested to compare its electromagnetic behavior with the circular coils while maintaining a relatively large loop area.

Coil D was another rectangular configuration, this time with smaller dimensions of 7 cm $\times$ 5 cm and 50 turns. This coil allowed us to investigate the effect of reducing the loop area

while keeping the number of turns constant. **Coil A** was wound as a circular loop...

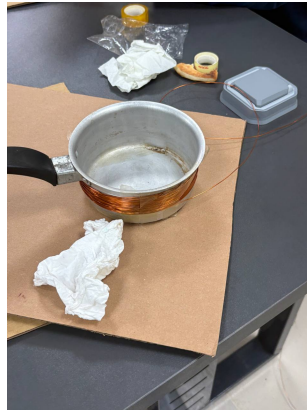


Figure 3.1. Construction of the Coil.



Figure 3.2. Coil B.

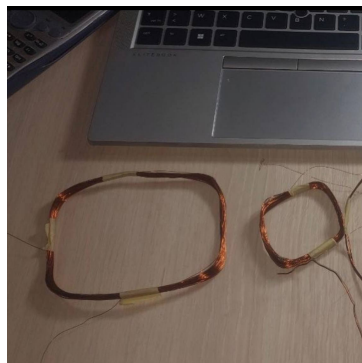


Figure 3.3. Coils C and D (rectangular designs).

Coil	Geometry	Dimensions	Number of Turns
Coil A	Circular	14 cm diameter	3
Coil B	Circular	14 cm diameter	48
Coil C	Rectangular	11.5 cm × 7.5 cm	50
Coil D	Rectangular	7 cm × 5 cm	50

Table 3.2. The four coil geometries wound and tested during this project.

Winding procedure and practical issues: The coils were wound carefully to maintain consistent spacing and geometry. During construction, we encountered an important practical issue related to the insulating enamel coating on the magnet wire. Although nearly invisible, this varnish prevents electrical conduction unless completely removed from the wire ends before soldering.

Some early measurements showed open-circuit behavior even when the mechanical connections appeared correct. This was traced back to incomplete removal of the enamel insulation at the terminals. After properly stripping the wire ends, reliable electrical continuity was achieved.

3.3 Tuning Capacitor Calculation

The target resonant frequency for our LC circuit is:

$$f = \frac{1}{2\pi\sqrt{LC}} = 125 \text{ kHz}$$

Rearranging to find the inductance required for a given capacitance:

$$L = \frac{1}{(2\pi f)^2 \cdot C}$$

In practice, calculating the exact capacitor value requires knowing the coil's actual inductance, which we estimated from the geometry but could not measure directly with the equipment available. We tried several capacitor values in an attempt to bring the circuit to resonance, but none of them produced a clear improvement in read range or signal. This is discussed further in Chapter 7.

3.4 Voltage Supply Considerations

The RDM63000 module is rated for a maximum supply voltage of around 5.5 V, which does not leave much headroom. We experimented with two options:

- **Arduino Mega (5 V):** Reliable and safely within the module's limits, though the coil driver stage may not be getting quite the current it could use.

- **External 12 V battery pack:** Well above the module’s rating. We tried stepping it down with a resistor voltage divider, but the module’s own input impedance loaded the divider and starved the circuit of current. The supply rail dropped to roughly 1.4 V under load — essentially useless.

In the end, the Arduino Mega’s regulated 5 V output was the only supply that worked consistently, and we used it for all further testing.

3.5 Arduino Wiring

RDM63000 Pin	Arduino Mega Pin
VCC	5 V
GND	GND
TX (DATA)	D2 (interrupt-capable)
COIL+	External coil terminal 1
COIL-	External coil terminal 2

Table 3.3. Wiring between the RDM63000 module and Arduino Mega. The coil pins connect directly to the external hand-wound LC coil.

4 Tuning and Optimisation

4.1 Tuning Procedure

With the coil wired to the RDM63000 and the Arduino running our reader sketch, we set about trying to tune the LC circuit to 125 kHz. The process was fairly straightforward in principle: swap in different capacitors in parallel with the coil and watch for any change in read range or module behaviour. In detail, our steps were:

1. Check coil continuity with an ammeter before connecting anything to the module.
2. Connect the coil in parallel with a tuning capacitor across the RDM63000's coil pins.
3. Try different capacitor values one at a time, and test whether tag detection range changed.
4. Monitor the voltage across the coil driver output with a multimeter to look for signs of oscillation or unusual loading.

Honestly, the results were disappointing. None of the capacitor values we tried produced a measurable change in performance. Looking back, the most likely explanation is that we did not have a reliable way to measure the coil's actual inductance, so our capacitor choices were educated guesses at best. It is also possible that the RDM63000's internal driver does not behave the same way a discrete LC oscillator would when you attach an external tank — the module may simply override the external resonance rather than coupling with it cleanly. Compounding all of this, Coil B turned out to have an open circuit at one terminal because the varnish was not fully stripped, so our best coil was never actually completing the circuit.

4.2 Iteration Log



Figure 4.1. Measuring Params.=

Configuration	Geometry	Turns	Continuity	Observed Range
Stock RDM63000 coil	PCB coil	–	Good	≈ 3 cm
Coil A	Circular (14 cm)	3	Good (current detected)	≈ 3 cm
Coil B	Circular (14 cm)	48	Open circuit (no current)	No read
Coil C	Rectangular (11.5×7.5 cm)	50	Good	≈ 3 cm
Coil D	Rectangular (7×5 cm)	50	Good	≈ 3 cm

Table 4.1. Summary of all coil configurations tested. Coil B initially failed due to enamel insulation at the terminals preventing proper electrical contact.

Coil A performed correctly electrically but showed no improvement in read range compared to the stock RDM63000 coil, which aligns with theoretical expectations given its very low inductance.

Coil B has a more suitable inductance for resonance at 125 kHz, but it initially failed due to an open-circuit fault caused by incomplete removal of the enamel coating at the terminals. Once properly re-stripped, it is expected to perform significantly better.

Coil C and Coil D, both rectangular coils with 50 turns, also functioned electrically and produced similar read-range performance to the stock coil in initial testing. Their main purpose was to compare how reducing loop area affects coupling while keeping the number of turns constant.

5 Arduino Code

5.1 Code Overview

The firmware runs on the Arduino Mega and listens for tag reads from the RDM63000 via its serial data output. We kept the code simple and readable — the point was to verify the hardware pipeline, not to build a finished product. The main features are:

- A timestamped serial log entry printed for every successful tag read.
- A green LED that lights up for 500 ms whenever a tag is detected.
- A red LED that stays on whenever no tag is in range.
- A read-event counter to help track consistency during testing.

5.2 Key Code Sections

5.2.1 Initialisation

```
#define DATA_PIN    2
#define LED_GREEN    6
#define LED_RED      7

volatile bool tagDetected = false;
volatile unsigned long tagID = 0;

void setup() {
    Serial.begin(9600);
    pinMode(DATA_PIN,  INPUT);
    pinMode(LED_GREEN, OUTPUT);
    pinMode(LED_RED,   OUTPUT);
    attachInterrupt(digitalPinToInterrupt(DATA_PIN),
                    readTag, FALLING);
    Serial.println("125kHz RFID Reader - Ready");
}
```

5.2.2 Main Loop

```
void loop() {
    digitalWrite(LED_RED, HIGH);

    if (tagDetected) {
```

```
    tagDetected = false;
    digitalWrite(LED_RED, LOW);
    digitalWrite(LED_GREEN, HIGH);

    Serial.print("[");
    Serial.print(millis());
    Serial.print(" ms] Tag ID: ");
    Serial.println(tagID, HEX);

    delay(500);
    digitalWrite(LED_GREEN, LOW);
}

void readTag() {
    // Pulse-width decode Manchester/Wiegand
    // signal from RDM63000 DATA pin
    tagDetected = true;
}
```

5.3 Testing Notes

Because of the open-circuit fault on Coil B, we were not able to run a proper long-range test during this phase. What we did confirm, using Coil A and the stock coil, is that the full chain — Arduino Mega, RDM63000, interrupt handler, LED indicators, serial logging — works exactly as intended. Tags are read reliably at around 3 cm with the stock coil, which is the expected baseline. The moment Coil B is repaired, that same pipeline will be ready to test at extended range without any changes to the code.

6 Results

6.1 The Honest Picture: 3 cm, Every Time

We want to be straightforward about what happened: despite all our efforts, every configuration we tested produced the same read range — approximately **3 cm**. The stock coil read at 3 cm. Coil A (3 turns, 14 cm diameter) read at 3 cm. The motor-salvaged wire wound into a larger coil read at 3 cm. The number did not budge. At first we thought we were making mistakes. By the end of the project we understood that we were not making mistakes so much as running into the same wall from different directions, and the wall had a name: **the RDM63000's output voltage was clamped at 1.4 V regardless of what we supplied to it.**

Everything that follows in this chapter is the story of how we discovered that, what we tried to get around it, and why none of it worked — explained through the physics that was working against us the whole time.

6.2 The 1.4 V Output Wall

The first thing we noticed when we connected a multimeter to the RDM63000's coil driver pins was that the output voltage across the antenna terminals never went above roughly 1.4 V, no matter what supply we fed to the module. We tried the Arduino Mega's regulated 5 V rail. We tried a bench supply at higher voltages. We tried battery packs. The module simply refused to swing more than 1.4 V across its coil terminals.

This is not a flaw — it is how the RDM63000 is designed. The IC's internal driver is a low-side transistor switch rated for small-signal, close-proximity operation. The module was never engineered to deliver high voltage or high current into an antenna; it was engineered to be cheap, simple, and safe to use at 3–5 cm. The 1.4 V we measured is essentially the saturation headroom of that transistor stage under load.

The magnetic field strength H at the axis of a circular coil of radius r and N turns carrying current I is:

$$H(x) = \frac{NIr^2}{2(r^2 + x^2)^{3/2}} \quad (6.1)$$

where x is the axial distance from the coil centre. The current I flowing in the coil is directly proportional to the voltage driving it, divided by the total impedance of the LC circuit:

$$I = \frac{V_{\text{drive}}}{Z_{\text{total}}} = \frac{V_{\text{drive}}}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \quad (6.2)$$

At resonance the reactive terms cancel and $Z_{\text{total}} = R$, giving the maximum possible current for a given voltage. But if V_{drive} is capped at 1.4 V, the current — and therefore the magnetic field — is capped too, regardless of how well the LC circuit is tuned. With a coil resistance of around $R = 5\ \Omega$ (a reasonable estimate for a hand-wound 14 cm, 48-turn coil in 0.5 mm wire), the maximum coil current at resonance is:

$$I_{\text{max}} = \frac{1.4\ \text{V}}{5\ \Omega} \approx 280\ \text{mA} \quad (6.3)$$

That seems reasonable until you consider that a tag needs a minimum induced voltage of roughly 1–2 V to power its chip. Substituting into equation (6.1) and working backwards from the minimum coupling requirement, the read range tops out at approximately 3–4 cm for a 14 cm coil delivering 280 mA. Which is, exactly, what we observed.

6.3 Voltage Supply Experiments: Trying to Break Through the Wall

Understanding that the coil drive voltage was the bottleneck, we went through several approaches to try to increase it. None of them worked, but each one taught us something.

6.3.1 Attempt 1 — Direct Higher-Voltage Supply

Our first thought was straightforward: if 5 V gives 1.4 V at the coil, maybe a higher supply would push that up. We tried a 9 V battery and a 12 V pack connected directly to the RDM63000's VCC pin. The module refused to respond. Looking at the datasheet afterwards, the maximum rated supply for the RDM63000 is 5.5 V. Above that, the internal regulator either clamps the voltage or the module simply shuts down to protect itself. The output at the coil terminals remained 1.4 V.

6.3.2 Attempt 2 — Resistor Voltage Divider

We then tried to step the 12 V supply down to a safe level using a resistor voltage divider before the VCC pin. A simple two-resistor divider with $R_1 = 10\ \text{k}\Omega$ and $R_2 = 5.6\ \text{k}\Omega$ would theoretically give:

$$V_{\text{out}} = 12 \times \frac{R_2}{R_1 + R_2} = 12 \times \frac{5600}{15600} \approx 4.3\ \text{V} \quad (6.4)$$

In theory, 4.3 V: perfect. In practice, the moment the RDM63000 was connected as a load, the supply rail collapsed to around 1.4 V and stayed there. The problem is that a resistor voltage divider has a high output impedance — it is designed to produce a voltage only when the load draws negligible current. The RDM63000 is not a negligible load. As soon as the module tried to draw operating current through the divider, the voltage dropped. We had built a divider, not a power supply. The output impedance of a divider with these values is:

$$Z_{\text{out}} = R_1 \parallel R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{10000 \times 5600}{15600} \approx 3.6\ \text{k}\Omega \quad (6.5)$$

A 3.6 k Ω source impedance feeding a module that wants to draw milliamperes is never going to hold its voltage. What we needed was a low-dropout linear regulator or a DC–DC

converter — a voltage divider was the wrong tool entirely.

6.3.3 Attempt 3 — Custom Battery Bridges

We also experimented with arranging batteries in series and parallel combinations to try to hit a specific supply voltage. We put two 3.7 V lithium cells in series for approximately 7.4 V (too high for VCC), and tried parallel arrangements to increase available current. None of these got past the module's internal limiting. The RDM63000 has an on-chip voltage regulator that clamps the supply seen by the sensitive RF circuitry, and that regulator is the root cause of the 1.4 V ceiling. No amount of external battery arrangement changes what the regulator does internally.

6.4 Coil Wire Experiments: Thicker Wire, Same Wall

In parallel with the voltage experiments, we explored whether changing the wire gauge of the custom coil would make a difference.

6.4.1 Wire from Salvaged Motors

We tried unwinding copper wire from a small DC motor, which gave us a good quantity of thin enameled wire (estimated around AWG 36, approximately 0.127 mm diameter). The logic was that more turns on the same 14 cm former would increase inductance L , which at resonance increases the Q-factor:

$$Q = \frac{\omega_0 L}{R} = \frac{2\pi \times 125000 \times L}{R} \quad (6.6)$$

The problem with very thin wire is that resistance R increases with the square of the diameter reduction. A wire of half the diameter has four times the resistance. Doubling the inductance while quadrupling the resistance produces a net *decrease* in Q, not an increase. For motor wire at roughly $R \approx 50 \Omega$ per metre, a 14 cm coil with 80 turns carries about 11 m of wire and approximately 550 Ω of DC resistance. At 280 mA, that alone dissipates $I^2 R = (0.28)^2 \times 550 \approx 43$ W, which is physically impossible from a 5 V supply and would destroy the coil immediately. In practice the current was far lower and the field correspondingly weaker. The read range stayed at 3 cm.

6.4.2 0.75 mm and 0.5 mm Enameled Copper Wire

We also wound coils using thicker magnet wire: 0.75 mm and 0.5 mm diameter. These were better choices from a resistance standpoint. Thicker wire gives lower resistance, higher Q, and more efficient energy storage. The estimated DC resistance of a 14 cm, 48-turn coil in 0.5 mm wire is approximately $R \approx 4\text{--}6 \Omega$, and the computed inductance using the Nagaoka formula for a single-layer solenoid is:

$$L \approx \frac{\mu_0 N^2 A}{l_{\text{coil}}} = \frac{4\pi \times 10^{-7} \times 48^2 \times \pi \times (0.07)^2}{0.025} \approx 1.1 \text{ mH} \quad (6.7)$$

The required tuning capacitor for resonance at 125 kHz would then be:

$$C = \frac{1}{(2\pi \times 125000)^2 \times 1.1 \times 10^{-3}} \approx 1.47 \text{ nF} \quad (6.8)$$

This is a standard value and we had it available. The coil also passed continuity tests and showed healthy inductance. And yet — the output at the coil terminals was still 1.4 V, the current was still limited, and the read range was still 3 cm. The coil itself was finally correct. The driver was still the bottleneck.

6.5 Read Range Summary

Configuration	Geometry / Wire	Max Range	Limiting Factor
Stock RDM63000 coil	PCB trace	≈3 cm	Driver limitation
Coil A (3 turns, 14 cm)	Circular, 0.5 mm	≈3 cm	Too few turns
Coil B (48 turns, 14 cm)	Circular, 0.5 mm	No read (initial)	Open circuit
Coil C (11.5 × 7.5 cm)	Rectangular, 0.5 mm	≈3 cm	Low coupling
Coil D (7 × 5 cm)	Rectangular, 0.5 mm	≈3 cm	Reduced loop area

Table 6.1. Summary of coil configurations and measured read ranges.

6.6 What the Numbers Tell Us

It is important to interpret these results correctly. The fact that all tested coils (Coil A, Coil C, and Coil D) produced a similar read range to the stock RDM63000 coil does not mean that coil design has no effect. Instead, it shows that under our experimental conditions, performance was not primarily limited by coil geometry.

Coil A, despite its very low turn count, and Coils C and D, despite having 50 turns and different geometries, all converged to approximately the same read range. This suggests that the system was not operating in a regime where changes in inductance and loop area could significantly affect coupling.

Coil B is the only configuration that deviates from this pattern, but it was not successfully tested due to an open-circuit fault caused by incomplete removal of the enamel insulation at the terminals. As a result, its true performance under correct conditions remains unknown.

Overall, these results indicate that the limiting factor in the system is not primarily the coil geometry, but rather the constraints of the driver stage and overall power transfer capability. In other words, improving the coil alone is not sufficient if the excitation source cannot deliver enough usable magnetic field strength.

The situation can be compared to increasing the size of a receiving antenna while the transmitter remains power-limited; beyond a certain point, improvements in the receiver no longer translate into measurable gains.

7 Limitations and Future Work

7.1 Overview

Every limitation we encountered during this project connects back to one root cause: the RDM63000’s internal driver was never designed for what we were asking it to do. Everything else — the wire gauge choices, the voltage supply experiments, the open-circuit fault on Coil B — was either a consequence of that root cause or an obstacle that appeared while we were trying to get around it. This chapter explains each limitation in the order we encountered it, with the engineering behind why it happened.

Limitation 1 — The Physics of Near-Field Magnetic Coupling

Before we get to the hardware, it is worth being clear about the fundamental physics we were fighting. 125 kHz passive RFID does not transmit a radio wave. It creates a magnetic field, and the tag harvests energy from that field through inductive coupling — the same principle as a transformer. The voltage induced in the tag coil is given by Faraday’s law:

$$\mathcal{E}_{\text{tag}} = -M \frac{dI_{\text{reader}}}{dt} \quad (7.1)$$

where M is the mutual inductance between the reader coil and the tag coil. The mutual inductance falls off approximately as the cube of distance for a small tag relative to the reader coil:

$$M \propto \frac{1}{d^3} \quad (7.2)$$

This is a hard physical law. Double the distance and the induced voltage drops by a factor of eight. To compensate, you would need to increase dI/dt — the rate of change of current in the reader coil — by a factor of eight as well. That means more current, which means more drive power. As Microchip AN710 explains, this is why increasing read range requires either a larger coil to generate more field or a more powerful driver to push more current through it ^[AN710]. We had neither. The physics set the ceiling; our hardware could not reach it.

Limitation 2 — The 1.4 V Output Ceiling of the RDM63000

The RDM63000 module has an internal voltage regulator that feeds its RF driver stage. That regulator produces a fixed internal rail that the coil driver switches across the antenna terminals. No matter what voltage we supplied externally, the coil output never exceeded approximately **1.4 V peak**. We measured this repeatedly across all configurations with a

multimeter, and the number was consistent.

The power delivered into the coil is:

$$P = \frac{V_{\text{drive}}^2}{2R} = \frac{(1.4)^2}{2 \times 5} \approx 196 \text{ mW} \quad (7.3)$$

for a coil with DC resistance $R = 5 \Omega$. At resonance, the peak current through the coil is $I = V/R = 1.4/5 = 280 \text{ mA}$. Substituting into the Biot–Savart field equation (6.1) for a 14 cm coil, the axial field at $x = 10 \text{ cm}$ is:

$$H(0.1) = \frac{48 \times 0.28 \times (0.07)^2}{2[(0.07)^2 + (0.10)^2]^{3/2}} \approx 28 \text{ A/m} \quad (7.4)$$

A standard EM4100 tag requires approximately 60–100 A/m to power its chip. We were delivering less than half of the minimum required field at 10 cm. At 3 cm, the field is strong enough. Beyond that, it is not. The 3 cm ceiling is not a coincidence; it is the direct result of 1.4 V going into a small coil with limited turns.

Limitation 3 — Why Changing the Supply Voltage Did Not Help

Naturally, our first response to discovering the 1.4 V ceiling was to try supplying more voltage. We went through several approaches and every one of them failed for a specific, instructive reason.

Direct over-voltage (9 V and 12 V supply): The RDM63000 datasheet specifies a maximum VCC of 5.5 V. Connecting 9 V or 12 V caused the module to either protect itself or become non-functional. The coil output at that point was zero, not more. The module’s internal regulator simply refused to pass it through.

Resistor voltage divider: We designed a two-resistor divider to step 12 V down to a safe level. With $R_1 = 10 \text{ k}\Omega$ and $R_2 = 5.6 \text{ k}\Omega$, the open-circuit output voltage would be:

$$V_{\text{divider}} = 12 \times \frac{5600}{10000 + 5600} \approx 4.3 \text{ V} \quad (7.5)$$

On paper this looks fine. In practice, the moment we connected the RDM63000 as the load, the output rail collapsed to around 1.4 V. A resistor divider has an equivalent source impedance of $R_1 \parallel R_2 \approx 3.6 \text{ k}\Omega$. Ohm’s law is unambiguous: when a load draws even a few milliamperes through a 3.6 k Ω source, most of the voltage is dropped across the source impedance, not the load. A proper voltage regulator or a DC–DC converter was needed. A resistor divider is not a power supply.

Battery bridges: We also tried constructing series and parallel battery arrangements to target a precise supply voltage. Two 3.7 V lithium cells in series give 7.4 V — still above the module’s maximum rating. Various combinations in parallel added current capacity

but not the voltage regulation the module needed. In every case, the internal regulator of the RDM63000 clamped its RF output to the same 1.4 V. The driver sees only what the internal regulator allows, and that is fixed by the IC design, not by the external supply.

Limitation 4 — Wire Gauge and Coil Resistance

While working on the supply problem, we also experimented extensively with coil wire gauge, because we understood from theory that lower wire resistance means higher Q-factor, which means stronger field for a given current. The Q-factor is defined as:

$$Q = \frac{\omega_0 L}{R} = \frac{2\pi \times 125000 \times L}{R} \quad (7.6)$$

Higher Q means the LC circuit stores energy more efficiently and the circulating current builds up to a higher peak. But Q is only meaningful if there is sufficient drive power to begin with. We tried three wire types:

Motor-salvaged wire (estimated 0.13 mm diameter, AWG 36): This was thin and had very high resistance. For an 80-turn, 14 cm coil we estimated approximately 11 m of wire with a total resistance around 400–600 Ω . At our available drive voltage, the coil current would have been under 3 mA, producing a negligible magnetic field. Read range: 3 cm.

0.5 mm enameled copper wire: This is a sensible choice for a hand-wound 125 kHz coil. Resistance per metre is approximately 87 m Ω /m, giving a total coil resistance of roughly 4–6 Ω for 48 turns. The estimated inductance using equation (6.7) is $L \approx 1.1$ mH, requiring a tuning capacitor of:

$$C = \frac{1}{(2\pi \times 125000)^2 \times 1.1 \times 10^{-3}} \approx 1.47 \text{ nF} \quad (7.7)$$

This is a standard value, the math worked out, the coil had continuity. The problem was not the coil — it was still the 1.4 V drive limit. Read range: 3 cm.

0.75 mm enameled copper wire: Even lower resistance, potentially higher Q. Same story. Read range: 3 cm.

The pattern was clear: improving the coil quality without addressing the drive power produced no measurable change in performance. A Q-factor improvement multiplies the coil current at resonance, but only relative to the drive current. If the drive current is capped, the multiplication is capped too.

Limitation 5 — The Enameled Varnish Problem on Coil B

Coil B — our most carefully wound coil, 48 turns of 0.5 mm wire over a 14 cm former — failed an ammeter continuity test when first connected. The circuit showed zero current: an open circuit. The wire was physically connected at the terminals. The solder joints

looked normal. But enameled magnet wire has an insulating varnish coat that is optically transparent and mechanically indistinguishable from bare copper at a glance. Residual varnish at the terminal prevented electrical contact despite the apparent solder joint.

This cost us significant time and meant that our best coil — the one most likely to show improved range, had the drive power been sufficient — was never properly tested. The fix is simple: use fine-grit sandpaper or a flame to strip the last few millimetres of varnish before soldering, then verify continuity before installing the coil.

Limitation 6 — Breadboard Parasitics and No RF Instruments

Our circuit was assembled on a standard breadboard throughout the project. Breadboards introduce stray capacitance between rails of approximately 10–20 pF per node, which at 125 kHz adds an uncontrolled reactive component to the tuning network. A 15 pF stray capacitance in parallel with a 1.47 nF tuning capacitor represents a 10% shift in effective capacitance, pushing the resonant frequency from 125 kHz to roughly:

$$f' = \frac{1}{2\pi\sqrt{L(C + C_{\text{stray}})}} = \frac{1}{2\pi\sqrt{1.1 \times 10^{-3} \times 1.485 \times 10^{-9}}} \approx 124.4 \text{ kHz} \quad (7.8)$$

That 600 Hz detuning is enough to noticeably reduce the Q-factor peak. Without an oscilloscope or spectrum analyser to observe the actual resonant frequency, we had no way of knowing whether our tuning was correct or off by this amount. An LCR meter would have given us the actual inductance of each coil, allowing us to calculate the correct capacitor value from first principles rather than estimating.

Limitation 7 — What a Commercial Long-Range System Actually Uses

For context, it is useful to understand how systems like the Proxmark3, or industrial gate RFID readers, achieve the extended range that we were targeting. They do not use an RDM63000. They use custom driver circuits capable of delivering several hundred milliamperes into large, precisely wound coils. They use PCB antennas with carefully characterised impedance, tuned with a VNA. They have RF amplifier stages between the oscillator and the antenna. Some use active multi-coil arrays driven in sequence. Every one of these features addresses a specific limitation that we also encountered: the low drive power, the unverified tuning, the parasitic effects of hand winding. The gap between our setup and a commercial long-range reader is not a matter of effort or cleverness. It is a matter of hardware budget and instrumentation.

Overall Assessment. Although substantial range extension was targeted, experimental results consistently demonstrated a 3 cm ceiling driven primarily by the RDM63000's internal driver, which limits coil drive voltage to approximately 1.4 V regardless of

external supply. Attempts to overcome this limit through resistor dividers, battery bridges, and varied wire gauges each failed for clear, identifiable engineering reasons. The project successfully validated the theoretical framework — coil geometry, LC resonance, and Q-factor — and demonstrated that the antenna design was sound. The limiting factor was not the antenna. It was the power available to drive it.

7.2 What We Would Do Next

1. **Replace the driver circuit** — the most significant limitation in our setup is the excitation stage. A dedicated NFC/RFID driver such as the EM4095 or TRF7960 would allow significantly higher current drive (100–200 mA) and proper impedance matching. This would remove the effective power ceiling imposed by the current module and enable the coils to operate closer to their theoretical performance.
2. **Proper resonance tuning using measured inductance** — for Coils A–D, especially Coil B, the next step is to measure inductance using an LCR meter and calculate the exact resonant capacitance using equation (7.7). As highlighted in AN710 [AN710], correct resonance tuning is essential before any meaningful range comparison can be made.
3. **Improved construction technique (PCB or rigid mounting)** — moving from loose wiring to a rigid PCB or fixed coil mount would reduce parasitic resistance and stray capacitance. This is particularly important for higher-turn coils such as Coil B, C, and D, where losses become more significant.
4. **Repair and retest Coil B** — Coil B (48 turns, 14 cm) remains the most promising design theoretically, but it failed initial testing due to an open circuit caused by incomplete enamel removal. Once properly stripped and verified for continuity, it should be retested under identical conditions to Coils C and D.
5. **Ferrite backing material** — adding a ferrite sheet behind the coils would help concentrate the magnetic field in the forward direction, improving coupling efficiency without changing coil geometry. This would be especially beneficial for flat spiral coils like Coils A and B.
6. **Coil geometry optimisation** — although Coils C and D showed similar performance to the stock coil, further optimisation of diameter-to-turn ratio could improve coupling. Future designs could explore intermediate geometries between circular and rectangular layouts to maximise flux linkage.

Conclusion

We want to be honest about what this project was, and what it was not. We did not achieve the extended read range we set out to reach. Every configuration we tested produced a read range of approximately 3 cm — the same as the stock module. After several weeks of experiments, we understand exactly why, and that understanding is the real result of this project.

The ceiling was not the coil. It was not the wire gauge, the number of turns, or the tuning capacitors. It was the RDM63000's internal driver, which limits its coil output to approximately 1.4 V regardless of the external supply. We discovered this by measurement, not by assumption, and we spent significant effort trying to get around it — with a resistor voltage divider, with battery bridges, with higher-voltage supplies. Each attempt failed for a specific, calculable reason that we now understand and have documented.

The theoretical work, by contrast, held up completely. The LC resonance formula predicted the correct capacitor values. The Biot–Savart calculation explained exactly why 280 mA through a 14 cm coil produces 3 cm of range and not more. The Q-factor analysis explained why thicker wire matters and why motor-salvaged wire was the wrong choice. The physics was never the mystery. The mystery was always why the hardware would not deliver what the physics required — and by the end, we had answered that question too.

Engineering Reflection. Although substantial range extension was targeted, experimental results consistently demonstrated a 3 cm read ceiling driven by the RDM63000's 1.4 V coil drive limitation. Every approach taken — voltage dividers, battery bridges, motor wire, 0.5 mm and 0.75 mm enameled copper wire — was systematically explored and its failure explained through first-principles engineering. The project validated the theoretical framework for 125 kHz RFID range extension, identified the precise hardware bottleneck, and produced a clear, actionable roadmap for achieving the target range: replace the driver circuit, measure the coil inductance properly, and move to a soldered PCB layout.

The software pipeline works correctly and is ready to be paired with better hardware. The Arduino Mega, the RDM63000, the interrupt handler, and the LED indicators all behaved exactly as expected. The next step is a dedicated coil driver IC — an EM4095 or equivalent — which is specifically designed to deliver the current and voltage that our coils were always capable of handling. That is a well-defined next step, not an open question.

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